Adaptive Cyber Defense Using Machine Learning-Based Cyber Threat Analysis of Multi-Source VM Telemetry

Project Synopsis Submitted

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by

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Objective:

The objective of this project is to design and implement a lightweight adaptive cyber defense framework that combines isolated virtual machine environments, controlled attack simulations, multi-source system telemetry, and machine learning-based behavioral analysis. The proposed system will study how sequences of otherwise ordinary system events can be correlated to identify suspicious activity and generate defensive recommendations.

The system aims to address the challenge of identifying multi-stage cyber threats that may not be distinguishable from individual log entries by providing the following capabilities:

1. **Controlled Attack Simulation:** Generate safe, reproducible attack-like activities inside an isolated virtual machine containing synthetic sensitive assets and controlled services.
2. **Multi-Source Telemetry Collection:** Collect authentication, process, file, privilege, and network-related events and convert them into a common structured representation.
3. **Machine Learning-Based Analysis:** Detect deviations from normal behavior and analyse temporal combinations of events to discover recurring suspicious attack patterns.
4. **Defensive Recommendation:** Map identified patterns to practical security controls such as improved auditing, access restrictions, monitoring rules, or configuration changes.
5. **Web-Based Monitoring:** Provide a dashboard through which users can inspect telemetry, anomalies, detected patterns, confidence/risk indicators, and recommendations.

Through this project, we intend to demonstrate a reproducible and cost-free approach to cybersecurity experimentation in which simulated adversarial behavior can be observed, analysed, and translated into defensive intelligence using machine learning.

Scope:

This project focuses on developing a lightweight cyber defense and threat-analysis framework using an isolated virtual machine and machine learning techniques. The implementation will be designed around free and open-source software and free-tier resources wherever applicable, with the primary development and experimentation environment being an 8 GB Apple Silicon M2 MacBook.

The virtual environment will contain synthetic sensitive files and controlled services. Normal system activity and predefined attack-like scenarios will be generated to produce representative telemetry. The collected events will be normalized, stored, and transformed into features suitable for machine learning and sequence analysis.

The machine learning component will primarily focus on anomaly detection and behavioral/temporal pattern analysis. Rather than developing general-purpose offensive malware or targeting real systems, the project will use controlled simulations to reproduce observable behaviors such as authentication anomalies, unusual process activity, privilege-related events, protected-file access, and abnormal network activity.

A web-based interface will be developed to provide an accessible view of the experiments and their results. The project will not use real classified or confidential information, will not target unauthorized systems, and will keep the experimental environment isolated. All sensitive-looking assets used in the project will be synthetic.

Need for the Application:

Modern cyber attacks are often multi-stage processes in which an adversary's activity is distributed across authentication, process execution, privilege changes, file access, and network communication. An individual event, when viewed in isolation, may appear harmless or ambiguous. However, the temporal sequence and correlation of several events can provide stronger evidence of suspicious behavior.

Conventional monitoring approaches frequently depend on predefined rules or signatures. Although such methods remain useful, they can struggle when an attack produces a new combination of individually normal events. Correlating multiple telemetry sources can provide a more complete representation of system behavior and create opportunities for machine learning to identify deviations and recurring patterns.

There is therefore a need for a controlled platform in which security events can be generated repeatedly, studied as sequences rather than isolated entries, and translated into actionable defensive recommendations. Such a platform can support cybersecurity learning and experimentation without exposing real systems or data to potentially harmful activity.

Project Description:

Problem Statement:

Cyber attacks frequently involve multiple stages distributed across different system components. Authentication events, process creation, privilege changes, file operations, and network connections may each appear legitimate when considered independently. As a result, analysing logs as isolated events can produce incomplete threat detection and may fail to reveal the overall behavioral pattern of an attack.

Furthermore, fixed rules and signatures generally detect behavior that has already been explicitly described. They may not effectively capture new combinations or temporal relationships between events. A system is therefore required that can combine multiple sources of virtual machine telemetry, establish a representation of normal activity, identify anomalous behavior, and analyse sequences of events to discover recurring suspicious patterns.

The proposed project addresses this problem by creating an isolated virtual environment in which normal and controlled attack-like activity can be generated repeatedly. The resulting telemetry will be processed using machine learning techniques, and the identified patterns will be converted into security findings and defensive recommendations that can be reviewed through a web-based interface.

Proposed Solution:

The proposed solution is to develop an adaptive cyber defense framework consisting of an isolated virtual machine environment, a controlled activity/attack simulation module, a multi-source telemetry collector, a feature and data-processing pipeline, a machine learning analysis module, a defensive recommendation layer, and a web-based dashboard.

A virtual machine will be configured with synthetic sensitive assets and controlled services. A baseline of normal activity will first be established. Controlled scenarios will then generate attack-like sequences, including combinations of authentication anomalies, process activity, privilege-related events, protected file access, and network events. The objective is to create repeatable experiments rather than to compromise real systems.

Telemetry from different sources will be normalized into a common event schema containing information such as timestamp, virtual machine identifier, event type, process or user context, resource involved, and source. Features derived from these events will be used by an anomaly detection component to identify deviations from normal activity.

A second analysis stage will examine sequences and combinations of events to identify recurring behavioral patterns. For example, a sequence involving an authentication anomaly followed by a privilege-related event, protected-file access, and unusual network activity may be assigned a higher risk than any individual event considered independently.

When a suspicious pattern is identified, the system will associate it with defensive recommendations. Recommendations will be generated from a controlled set of security responses and may include increased auditing, access restrictions, process monitoring, or additional detection rules. The web dashboard will present the detected pattern, supporting events, confidence/risk information, and recommended controls.

The overall workflow will follow the cycle: **Simulate → Collect Telemetry → Analyse → Identify Pattern → Recommend Defense**. The implementation will remain within isolated environments and will use synthetic data so that the complete system can be demonstrated safely and without paid infrastructure.

Functionalities to be Implemented:

- Create and configure an isolated ARM64-compatible virtual machine environment for repeatable cybersecurity experiments.
- Generate normal system activity and controlled attack-like scenarios using synthetic sensitive files and services.
- Collect authentication, process, file, privilege, and network-related telemetry from the virtual machine.
- Normalize heterogeneous logs into a structured event schema and store experiment data for analysis.
- Build a baseline representation of normal virtual machine behavior.
- Detect anomalous activity using lightweight machine learning-based anomaly detection.
- Analyse temporal sequences and combinations of events to identify recurring suspicious behavioral patterns.
- Assign confidence or risk indicators to identified patterns based on the observed telemetry.
- Generate defensive recommendations mapped to detected behavioral patterns.
- Provide a web-based dashboard for logs, experiment summaries, anomalies, patterns, and recommendations.
- Maintain experiment identifiers and results so that different scenarios can be compared.
- Evaluate recommendations through controlled comparison of observed telemetry and detection outcomes where feasible.
- Keep all experiments isolated and ensure that no real classified, confidential, or unauthorized data or infrastructure is involved.

Hardware Requirements:

- **Processor:** Apple Silicon M2 or equivalent modern processor.
- **RAM:** Minimum 8 GB; the implementation will be optimized for an 8 GB MacBook Air by running experiments sequentially and keeping the virtual machine lightweight.
- **Storage:** Approximately 20–30 GB of available SSD storage for the guest operating system, datasets, logs, source code, and experiment snapshots.
- **GPU:** Not required. The proposed baseline machine learning techniques are intended to run on CPU resources.
- **Network:** Stable Internet connection for downloading open-source packages, Linux images, documentation, and optional free-tier resources.
- **Primary Development System:** Apple MacBook Air with M2 processor and 8 GB unified memory.

The project will avoid dependence on paid cloud infrastructure. If a free-tier remote VM is used for an experiment, the same core workflow will remain reproducible locally so that the project is not dependent on a paid service.

Software Requirements:

- **Operating System:** macOS on Apple Silicon.
- **Virtualization:** UTM/QEMU or another free ARM64-compatible virtualization solution.
- **Guest Operating System:** Ubuntu or Debian ARM64 Linux.
- **Programming Language:** Python.
- **Machine Learning:** scikit-learn for anomaly detection and baseline models; PyTorch may be considered for advanced sequence analysis if required.
- **Data Processing:** pandas and NumPy.
- **Telemetry:** Linux audit/journald facilities together with lightweight custom collectors.
- **Database:** SQLite for lightweight local experiment storage.
- **Backend:** FastAPI or Flask.
- **Web Dashboard:** Streamlit or a lightweight JavaScript/TypeScript frontend.
- **Visualization:** Matplotlib and/or Plotly.
- **Version Control:** Git.
- **Development Environment:** Visual Studio Code and Jupyter Notebook.
- **Optional Cloud Resources:** Only free-tier resources, if used. The project will not require paid cloud infrastructure to demonstrate its core functionality.
- **Web Browser:** Chrome, Firefox, Safari, or another modern browser for accessing the dashboard.

All primary components will be selected from free and open-source alternatives. The system architecture will be kept lightweight so that the complete prototype remains achievable with the available hardware and without recurring infrastructure costs.

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